Robust extended Kalman smoothing under model uncertainties

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***Abstract*—This paper proposes a robust extended Kalman smoothing approach in the case that there is a modeling mismatch between the given nominal nonlinear state-space model and the actual one. The resulting algorithm follows a linearization process and takes the form of a robust fixed-lag smoother. At each time step the computation of the smoothing gain of the linearized model is required. The latter is derived from a dynamic game: one player selects the least favorable model in a prescribed "ball" centered about the nominal linearized model in Kullback-Leibler (KL) divergence topology, and the other selects the optimal smoothing gain. Finally, some numerical experiments show the effectiveness of the proposed algorithm.**

***Index Terms*—fixed-lag smoothing, extended Kalman filter, model uncertainty**

I. Introduction

Fixed-lag Kalman smoothers (FKS) are widely used in various applications, in particular to communication systems [1] and to some control or tracking scenarios [2]. In those applications, indeed, the introduction of a delay is acceptable to yield a helpful improvement in the estimation of the variables of interest using a fixed-lag smoother. In fact, it is well-known that FKS can be interpreted as a Kalman filter (KF) corresponding to an augmented state space model [3], and thus FKS can be implemented recursively. However, in practice, the actual model is known imprecisely and only the nominal model is known. This is due to the following two reasons. First, the physical models have a natural description using continuous time models and thus the nominal model will be different from the actual one because of its discrete-time nature. Second, the nominal model is often an inaccurate version of the actual model because the noise processes are not precisely known.

One possible way to address model uncertainty is to take a risk sensitive philosophy that is the optimal filter minimizes an exponential loss function which severely penalizes large errors [4]–[6]. Then, it has been proved that such filters are the solution of a minimax game [7], [8]. In particular, an incremental risk sensitive paradigm has been considered in [9]: roughly speaking the model uncertainty cannot be concentrated only in one time step, but rather spread over the time. Later on, various extensions of this robust filtering approach have been proposed such as different ways to quantify model uncertainty [10], [11], the case of non-reachable state space models [12], [13], the distributed case [14], [15], and the fixed-lag smoothing problem [16], [17]. In particular, in [16] we have designed a robust fixed-lag smoother in the case the model is linear. However, in many practical situations, systems are characterized by nonlinearities. All the aforementioned algorithms, however, only work in the linear case. The extended Kalman filter (EKF) [18] generalizes KF in the nonlinear setting: the main idea is to linearize the nonlinear model around the previous estimates and adapt the KF recursions for the linearized model. The risk sensitive version, called robust extended Kalman filter, has been studied recently in [19]: in such a setting the nominal model is nonlinear. The smoothing problem in the nonlinear case has been also addressed in [20], [21]. Hereafter, robust nonlinear smoothers have been also studied in the recent literature, in particular for the heavy-tailed process noise [22], non-Gaussian process and measurement noises [23], or even their combination [24]. However, an incremental risk-sensitive approach to solve the fixed-lag smoothing problem under model uncertainties is still missing.

The contribution of this paper is to propose an incremental risk sensitive fixed-lag extended Kalman smoother (REKS), which combines the ideas from the robust fixed-lag smoothing problem in [16] and the robust extended Kalman filtering problem in [19]. This problem could appear trivial because a fixed-lag smoothing problem can be cast as a filtering problem, through an augmented state space model, and thus REKS seems just a special instance of the approach proposed in [19]. However, this is not true. Indeed, a fundamental hypothesis in [19] is that the noise affects all components of the state equation, but such an assumption does not hold here because in the augmented state space we will consider there are some state components which are related in a deterministic way.

The outline of the paper is as follows. Section II is devoted to the problem formulation. In Section III, we derive the extended robust Kalman smoother. Then, some numerical experiments are presented in Section IV. Finally, Section V regards the conclusions and future research directions.

II. Problem Formulation

We consider the following nominal nonlinear state-space model:

$$\begin{aligned} x_{t+1} &= f(x_t) + Bv_t \\ y_t &= h(x_t) + Dv_t \end{aligned} \tag{1}$$

where $B \in \mathbb{R}^{n\times(n+p)}$, $D \in \mathbb{R}^{p\times(n+p)}$ and $v_t \in \mathbb{R}^{n+p}$ is white Gaussian noise (WGN) with unit variance; $x_t \in \mathbb{R}^n$ is the state, $y_t \in \mathbb{R}^p$ is the output; functions $f(\cdot)$ and $h(\cdot)$ are assumed to be known and differentiable. Without loss of

generality we assume that B and D are full row rank and such that $BD^\top = 0$. In plain words, we assume that the noise processes $u_t := Bv_t$ and $w_t := Dv_t$ acting on the state and measurement equations, respectively, are full rank and independent.

Consider the following fixed-lag smoothing problem: find an estimate $\hat{x}_{t-L+1|t}$, recursively, of the state x_{t-L+1} given $Y_t = \{y_0, \cdots, y_t\}$ where $L > 0$ is the lag. The latter can be treated as an "augmented" filtering problem [20], see also the seminal work [25] for the linear case. More precisely, [20] considers the following augmented nominal model:

$$\begin{aligned} \xi_{t+1} &= \bar{f}(\xi_t) + \bar{B}v_t \\ y_t &= \bar{h}(\xi_t) + \bar{D}v_t \\ x_{t-L+1} &= \bar{H}\xi_{t+1} \end{aligned} \tag{2}$$

where the augmented state $\xi_t \in \mathbb{R}^{(L+1)n}$, $\bar{B} \in \mathbb{R}^{(L+1)n\times(p+n)}$, $\bar{D} \in \mathbb{R}^{p\times(p+n)}$ and $\bar{H} \in \mathbb{R}^{n\times(L+1)n}$ are such that

$$\begin{aligned} &\xi_t = \begin{bmatrix} x_t^\top & x_{t-1}^\top & \cdots & x_{t-L}^\top \end{bmatrix}^\top,\ \bar{D} = D, \\ &\bar{B} = \begin{bmatrix} B^\top & 0 & \cdots & 0 \end{bmatrix}^\top,\ \bar{H} = \begin{bmatrix} 0 & 0 & \cdots & I_n \end{bmatrix}. \end{aligned}$$

Moreover, the augmented functions $\bar{f}(\xi_t)$ and $\bar{h}(\xi_t)$ are defined as follows:

$$\bar{f}(\xi_t) = \begin{bmatrix} f(x_t)^\top & x_t^\top & \cdots & x_{t-L+1}^\top \end{bmatrix}^\top,\ \bar{h}(\xi_t) = h(x_t).$$

It is worth noting that $\bar{f}$ and $\bar{h}$ are differentiable by construction. As pointed out in [20], the fixed-lag smoothing problem can be transformed into a filtering problem through the augmented model (2). Let $\hat{x}_{t-l+1|t}$ be the smoothed estimate of x_{t-L+1} given Y_t. Moreover, let $\hat{\xi}_{t|t}$ and $\hat{\xi}_t$ represent the estimate of ξ_t given Y_t and Y_{t-1}, respectively. Adopting the philosophy of the extended Kalman filter (EKF) [26], at time t we linearize the two augmented equations in (2) around the previous estimates $\hat{\xi}_{t|t}$ and $\hat{\xi}_t$, respectively:

$$\begin{aligned} \xi_{t+1} &= \bar{A}_t\xi_t - \bar{A}_t\hat{\xi}_{t|t} + \bar{f}(\hat{\xi}_{t|t}) + \bar{B}v_t \\ y_t &= \bar{C}_t\xi_t - \bar{C}_t\hat{\xi}_t + \bar{h}(\hat{\xi}_t) + \bar{D}v_t \end{aligned} \tag{3}$$

where

$$\bar{A}_t = \left.\frac{\partial \bar{f}(\xi)}{\partial \xi}\right|_{\xi=\hat{\xi}_{t|t}},\ \bar{C}_t = \left.\frac{\partial \bar{h}(\xi)}{\partial \xi}\right|_{\xi=\hat{\xi}_t}.$$

Therefore, in view of (2), the a priori estimate $\hat{\xi}_{t+1}$ of ξ_{t+1} given Y_t is obtained by

$$\begin{aligned} \hat{\xi}_{t+1} &= \bar{f}(\hat{\xi}_{t|t}) \\ \hat{x}_{t-L+1|t} &= \bar{H}\hat{\xi}_{t+1} \end{aligned} \tag{4}$$

while the a posteriori estimate $\hat{\xi}_{t+1|t+1}$ of ξ_{t+1} given Y_{t+1} is given by the Kalman prediction step according to the linearized model (3):

$$\hat{\xi}_{t|t} = \hat{\xi}_t + G_t(y_t - \bar{h}(\hat{\xi}_t)) \tag{5}$$

where G_t is the Kalman filtering gain. In practice, physical models have a natural description using continuous time models and thus the nominal model (1) will be different from the actual one because of its discrete-time nature. On the other hand, the nominal model is often an inaccurate version of the actual model because the noise processes are not precisely known, e.g. the latter are not necessarily Gaussian distributed and matrices B, D are estimated from the available data. Thus, taking into account the aforementioned model uncertainties, we seek to find a robust approach for estimating x_{t-L+1} given Y_t.

Next, we consider the situation in which the actual model is different from the nominal one in (1), or equivalently (2). Let $z_t = [\xi_{t+1}^\top\ y_t^\top]^\top$. Then, the linearized nominal model (3) can be characterized by the nominal Gaussian transition probability of z_t given ξ_t as follows:

$$\bar{\phi}_t(z_t|\xi_t) \sim \mathcal{N}\left(\bar{m}_{z_t|\xi_t}\ ,\ \bar{K}_{z_t|\xi_t}\right), \tag{6}$$

where

$$\begin{aligned} \bar{m}_{z_t|\xi_t} &= \begin{bmatrix} \bar{A}_t\xi_t - \bar{A}_t\hat{\xi}_{t|t} + \bar{f}(\hat{\xi}_{t|t}) \\ \bar{C}_t\xi_t - \bar{C}_t\hat{\xi}_t + \bar{h}(\hat{\xi}_t) \end{bmatrix}, \\ \bar{K}_{z_t|\xi_t} &= \begin{bmatrix} \bar{B} \\ \bar{D} \end{bmatrix}\begin{bmatrix} \bar{B}^\top & \bar{D}^\top \end{bmatrix}. \end{aligned}$$

Note that $\bar{K}_{z_t|\xi_t} \in \mathbb{R}^{(L+1)n+p\times(L+1)n+p}$ is singular since we have $\operatorname{rank}(\bar{B}\bar{B}^\top) = n$ and thus $\operatorname{rank}(\bar{K}_{z_t|\xi_t}) = n+p$. Hence, $\bar{\phi}_t(z_t|\xi_t)$ is a degenerate probability density function and its support is given by the $n+p$-dimensional affine subspace denoted by $\bar{\mathcal{A}}_t$, i.e.,

$$\bar{\mathcal{A}}_t = \left\{\bar{m}_{z_t|\xi_t} + v, \quad v \in \operatorname{Im}(\bar{K}_{z_t|\xi_t})\right\}$$

where $\operatorname{Im}(K)$ denotes the image of matrix K.

Remark 1. *It is worth noting that the augmented noise process $\bar{u}_t := \bar{B}v_t$ affects only the first n state components of ξ_{t+1} in (2). Indeed, it is easy to see that the other Ln components can be obtained by ξ_t through a deterministic mapping. Thus, the robust extended Kalman filter proposed in [19] cannot be applied because it requires that the covariance matrix of process $\bar{u}_t$ is full rank and (as consequence) $\bar{\phi}_t$ is a non-degenerate probability density.*

In what follows we generalize the robust paradigm in [19] to the case in which we have a nominal model as the one in (2). We assume that the actual density $\tilde{\phi}_t(z_t|\xi_t)$ (not necessarily Gaussian distributed) of z_t given ξ_t has the same support of the nominal one. Moreover, we measure the mismatch between the actual model and the nominal one through the KL divergence between the probability densities $\tilde{\phi}_t$ and $\bar{\phi}_t$:

$$\begin{aligned} \mathcal{D}(\tilde{\phi}_t, \bar{\phi}_t) = \int_{\mathcal{A}_t}\int_{\bar{\mathcal{A}}_t} &\tilde{\phi}_t(z_t|\xi_t)\tilde{p}_t(\xi_t|Y_{t-1}) \\ &\times \log\left(\frac{\bar{\phi}_t(z_t|\xi_t)}{\tilde{\phi}_t(z_t|\xi_t)}\right) dz_t d\xi_t \end{aligned} \tag{7}$$

where $\tilde{p}_t(\xi_t|Y_{t-1})$ is the actual *a priori* probability density of ξ_t conditioned on the observations Y_{t-1} and $\mathcal{A}_t$ is the support of $\tilde{p}_t(\xi_t|Y_{t-1})$. In what follows, we assume that the actual probability density $\tilde{\phi}_t$ is within an ambiguity set, named $\mathcal{B}_t$, which is a "ball" centered about the nominal one $\bar{\phi}_t$:

$$\mathcal{B}_t = \left\{\ \tilde{\phi}(z_t|\xi_t)\ \text{s.t.}\ \mathcal{D}(\tilde{\phi}_t, \bar{\phi}_t) \le c_t\right\} \tag{8}$$

where $c_t > 0$, hereafter called the tolerance, measures the budget of model uncertainty between the nominal and the actual model at each time step t through the KL divergence. Note that the KL divergence is the natural metric to measure such a mismatch in the case that the nominal model is inferred from data, see [5].

Remark 2. *This type of model uncertainty arises when the nominal model is estimated from data, i.e. model (1) is selected from a parametric Gaussian model class according to the maximum likelihood principle. More details can be found in [12, pages 3459-3460]. Moreover, if we want to spread equally the model uncertainty for $t = 1 \dots N$ then we can take*

$$c_t \approx \frac{\ell}{\tilde{N}}$$

where ℓ is the value of the negative log-likelihood for the estimated model, in our case model (1), and $\tilde{N}$ is the length of the data.

Drawing inspiration from the robust philosophy in [19], we consider an "augmented" extended Kalman filter where the filtering Kalman gain comes from the following minimax problem:

$$\hat{\xi}_{t+1} = \underset{g_t \in \mathcal{G}_t}{\operatorname{argmin}} \max_{\tilde{\phi}_t \in \mathcal{B}_t} J_t(\tilde{\phi}_t, g_t) \tag{9}$$

where

$$\begin{aligned} J_t(\tilde{\phi}_t, g_t) =& \frac{1}{2}\tilde{\mathbb{E}}\left[\|\bar{H}\left(\xi_{t+1} - g_t\left(y_t\right)\right)\|^2 | Y_{t-1}\right] \\ =& \frac{1}{2}\int_{\mathcal{A}_t}\int_{\bar{\mathcal{A}}_t} \|\bar{H}\left(\xi_{t+1} - g_t\left(y_t\right)\right)\|^2 \tilde{\phi}_t\left(z_t|\xi_t\right) \\ & \times \tilde{p}_t\left(\xi_t|Y_{t-1}\right) dz_t d\xi_t, \end{aligned} \tag{10}$$

$\mathcal{G}_t$ denotes the class of estimators with finite second-order moments with respect to all the densities $\tilde{\phi}_t\left(z_t|\xi_t\right)\tilde{p}_t(\xi_t|Y_{t-1})$ such that $\tilde{\phi}_t \in \mathcal{B}_t$. Notice that $\tilde{\phi}_t$ must satisfy the constraint:

$$I_t(\tilde{\phi}_t) \triangleq \int_{\check{A}_t}\int_{\mathcal{A}_t} \tilde{\phi}_t\left(z_t|\xi_t\right)\tilde{p}_t\left(\xi_t|Y_{t-1}\right) dz_t d\xi_t = 1. \tag{11}$$

Remark 3. *The objective function in [19] is given by*

$$\frac{1}{2}\tilde{\mathbb{E}}\left[\|\xi_{t+1} - g_t\left(y_t\right)\|^2 | Y_{t-1}\right]$$

which is different from the one in (9). Indeed, here we introduce the weight matrix $\bar{H}$ to select only the n components corresponding to the ones of the original state in (1). Indeed, our final aim is not to construct an estimator for ξ_t, but rather a smoothed estimator for x_t.

In the next section our aim is to address the following robust smoothing problem:

Problem. *Design a robust fixed-lag extended Kalman smoother (REKS) under the model uncertainties described by the ambiguity set $\mathcal{B}_t$, for $t = 0 \dots N$, which exploits the minimax problem in (9) for the linearized model.*

III. Robust extended Kalman smoother

Consider the matrix-partition unfolding of $\hat{\xi}_t$ and $\hat{\xi}_{t|t-1}$:

$$\hat{\xi}_t := \left[(\hat{x}_{t|t-1})^\top (\hat{x}^1_{t|t-1})^\top \cdots (\hat{x}^j_{t|t-1})^\top \cdots (\hat{x}^L_{t|t-1})^\top\right]^\top, \tag{12}$$

$$\hat{\xi}_{t|t} := \left[(\hat{x}_{t|t})^\top (\hat{x}^1_{t|t})^\top \cdots (\hat{x}^j_{t|t})^\top \cdots (\hat{x}^L_{t|t})^\top\right]^\top, \tag{13}$$

where $\hat{x}^j_{t|t-1} = \hat{x}_{t-j|t-1}$, $j > 0$, is the estimate of x_{t-j} given Y_{t-1} and $\hat{x}^j_{t|t} = \hat{x}_{t-j|t}$, $j > 0$, is the estimate of x_{t-j} given Y_t. Then, the partitions of $\bar{A}_t$ and $\bar{C}_t$ are as follows:

$$\bar{A}_t = \left.\frac{\partial \bar{f}(\xi)}{\partial \xi}\right|_{\xi=\hat{\xi}_{t|t}} = \begin{bmatrix} A_t & 0 & \cdots & 0 \\ I & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & I & 0 \end{bmatrix}, \tag{14}$$

$$\bar{C}_t = \left.\frac{\partial \bar{h}(\xi)}{\partial \xi}\right|_{\xi=\hat{\xi}_t} = \begin{bmatrix} C_t & 0 & \cdots & 0 \end{bmatrix} \tag{15}$$

where

$$A_t = \left.\frac{\partial f(x)}{\partial x}\right|_{x=\hat{x}_{t|t}}, \quad C_t = \left.\frac{\partial h(x)}{\partial x}\right|_{x=\hat{x}_t}.$$

Notice that $\hat{x}_{t|t}$ and $\hat{x}_t$ can be easily computed from $\hat{\xi}_{t|t}$ and $\hat{\xi}_t$, respectively, as follows:

$$\hat{x}_{t|t} = \tilde{H}\hat{\xi}_{t|t}\ ,\ \hat{x}_t = \tilde{H}\hat{\xi}_t$$

with $\tilde{H} = \begin{bmatrix} I_n & 0 & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{n \times n(L+1)}$.

In addition, in view of [12, Lemma 2], we know that since the nominal density $\bar{\phi}_t$ shown in (6) is Gaussian, then the least favorable density $\tilde{\phi}_t$ is also Gaussian. However, the nominal and least favorable marginal densities

$$\begin{aligned} \bar{p}_t\left(z_t|Y_{t-1}\right) &:= \int_{\bar{\mathcal{A}}_t} \bar{\phi}_t\left(z_t|\xi_t\right)\bar{p}_t\left(\xi_t|Y_{t-1}\right) d\xi_t \\ \tilde{p}_t\left(z_t|Y_{t-1}\right) &:= \int_{\bar{\mathcal{A}}_t} \tilde{\phi}_t\left(z_t|\xi_t\right)\tilde{p}_t\left(\xi_t|Y_{t-1}\right) d\xi_t \end{aligned} \tag{16}$$

are not necessarily Gaussian because $\tilde{p}_t(\xi_t|Y_{t-1})$ is unknown. Indeed, the least favorable density solution to the minimax problem (9) depends on the estimator, while the optimal estimator depends on the least favorable density. Hence, in order to break this deadlock, we assume that the actual *a priori* probability density $\tilde{p}_t(\xi_t|Y_{t-1}) \sim \mathcal{N}(\hat{\xi}_t, \tilde{V}_t)$ with $\tilde{V}_t > 0$. By Lemma 1 and Lemma 2 in [16] it follows that the maximizer $\tilde{\phi}_t$ of (9) is such that $\tilde{p}_t\left(z_t|Y_{t-1}\right)$ is non-degenerate Gaussian density. Since $\tilde{p}_t\left(z_t|Y_{t-1}\right)$ enjoys the aforementioned properties and $\tilde{V}_t > 0$, by [16, Theorem 3] it follows that the optimal robust estimator solution to (9) is:

$$\hat{\xi}_{t+1} = \bar{A}_t\hat{\xi}_{t|t}, \tag{17}$$

where

$$\hat{\xi}_{t|t} = \hat{\xi}_t + \bar{G}_t(y_t - \bar{h}(\hat{\xi}_t))$$

and the robust filtering gain $\bar{G}_t$ now is given by:

$$\bar{G}_t = \tilde{V}_t\bar{C}_t^\top(\bar{C}_t\tilde{V}_t\bar{C}_t^\top + \bar{D}\bar{D}^\top)^{-1}$$

where

$$\tilde{V}_{t+1} = (\bar{P}_{t+1}^{-1} - \theta_t \bar{H}^\top \bar{H})^{-1}, \tag{18}$$

$$\begin{aligned}\bar{P}_{t+1} :=& \bar{A}_t \tilde{V}_t \bar{A}_t^\top - \bar{A}_t \tilde{V}_t \bar{C}_t^\top (\bar{C}_t \tilde{V}_t \bar{C}_t^\top + \bar{D}\bar{D}^\top)^{-1} \\ & \times \bar{C}_t \tilde{V}_t \bar{A}_t^\top + \bar{B}\bar{B}^\top. \end{aligned} \tag{19}$$

Moreover, $\theta_t > 0$, the so-called the risk sensitivity parameter, is the unique solution to $\gamma(\bar{P}_{t+1}, \theta_t) = c_t$ and such that (18) is well defined, i.e. $\tilde{V}_{t+1} > 0$. Such θ_t always exists. Here,

$$\begin{aligned}\gamma(\bar{P}, \theta) :=& \frac{1}{2} \left\{ \log\det(I - \theta H^\top H \bar{P}) \right\} \\ & + \frac{1}{2} \left\{ \mathrm{tr} \left[(I - \theta H^\top H \bar{P})^{-1} - I \right] \right\}\end{aligned}$$

where $\mathrm{tr}(P)$ is the trace of the matrix P and $\det(P)$ is the determinant of P. Thus, the estimator corresponding to the original nonlinear model is as the previous one where (17) is replaced by

$$\hat{\xi}_{t+1} = \bar{f}(\hat{\xi}_{t|t}).$$

Finally, we have:

$$\hat{x}_{t-L+1|t} = \bar{H}\hat{\xi}_{t+1}.$$

Remark 4. *The resulting robust estimator is similar to (4)-(5), however, the filtering gain $\bar{G}_t$ and the posterior covariance matrix of the estimation error $\tilde{V}_t$ are different.*

In view of the special structure of the augmented model (3) we have:

$$\bar{h}(\hat{\xi}_t) = h(\hat{x}_t)$$

and

$$\bar{f}(\hat{\xi}_{t|t}) = \begin{bmatrix} f(\hat{x}_{t|t}) \\ \check{H}\hat{\xi}_{t|t} \end{bmatrix}$$

where $\check{H} = \begin{bmatrix} I_{nL} & 0 \end{bmatrix} \in \mathbb{R}^{nL \times n(L+1)}$. The resulting REKS procedure is reported in Algorithm 1. It is not difficult to show, like in [16, Section 4] for the linear case, that the complexity of Algorithm 1 with respect to the instance L is $O(L^3)$. The presence of $\tilde{V}_t$, which requires (at Step 14) the computation of θ_t, is the key point to make our REKS work under model uncertainties. Note that the computation of the risk sensitivity parameter θ_t can be performed through the bisection method, since a closed form solution does not exist.

Remark 5. *An alternative version of Algorithm 1 would be the one with θ_t replaced by a constant risk sensitivity parameter $\theta > 0$ chosen by the user, and thus Step 13 is removed. The latter represents a generalization of the robust fix-lag smoother in [27] to the nonlinear case. However, the choice of θ is quite problematic: if θ is chosen too large the robust smoother does not exist (even in the linear case). The largest value of θ for which the smoother exists depends on the state space matrices B, D and $\{A_t, C_t,\ t = \dots N\}$. Since the sequence of matrices $\{A_t, C_t,\ t = \dots N\}$ is not known a priori because it depends on the particular linearization performed at the different steps, the choice of θ is extremely difficult. Instead, in our setup we do not have this problem.*

Algorithm 1 Robust Extended Kalman Smoother

Input $\hat{\xi}_0$, $\tilde{V}_0$, f, h, $\bar{B}$, $\bar{D}$, $\bar{H}$, $\check{H}$, c_t, L, $y_0 \dots y_N$
Output $\hat{x}_{t-L+1|t},\ t = L-1, \cdots, N$

1: **for** $t = 0 : N$ **do**
2: $\quad \hat{x}_t = \tilde{H}\hat{\xi}_t$
3: $\quad C_t = \frac{\partial h(x)}{\partial x}\Big|_{x=\hat{x}_t}$
4: $\quad$ *Build $\bar{C}_t$ using (14)*
5: $\quad \bar{G}_t = \tilde{V}_t \bar{C}_t^\top (\bar{C}_t \tilde{V}_t \bar{C}_t^\top + \bar{D}\bar{D}^\top)^{-1}$
6: $\quad \hat{\xi}_{t|t} = \hat{\xi}_t + \bar{G}_t (y_t - h(\hat{x}_t))$
7: $\quad \hat{x}_{t|t} = \tilde{H}\hat{\xi}_{t|t}$
8: $\quad A_t = \frac{\partial f(x)}{\partial x}\Big|_{x=\hat{x}_{t|t}}$
9: $\quad$ *Build $\bar{A}_t$ using (15)*
10: $\quad \hat{\xi}_{t+1} = \begin{bmatrix} f(\hat{x}_{t|t}) \\ \check{H}\hat{\xi}_{t|t} \end{bmatrix}$
11: $\quad \bar{P}_{t+1} = \bar{A}_t \tilde{V}_t \bar{A}_t^\top - \bar{A}_t \tilde{V}_t \bar{C}_t^\top$
12: $\qquad \times (\bar{C}_t \tilde{V}_t \bar{C}_t^\top + \bar{D}\bar{D}^\top)^{-1} \bar{C}_t \tilde{V}_t \bar{A}_t^\top + \bar{B}\bar{B}^\top$
13: $\quad$ *Find θ_t s.t. $\gamma(\bar{P}_{t+1}, \theta_t) = c_t$*
14: $\quad \tilde{V}_{t+1} = (\bar{P}_{t+1}^{-1} - \theta_t \bar{H}^\top \bar{H})^{-1}$
15: $\quad \hat{x}_{t-L+1|t} = \bar{H}\hat{\xi}_{t+1}$

IV. Numerical experiments

To evaluate the effectiveness of REKS proposed in Algorithm 1, we consider the following nominal discrete-time nonlinear state space model:

$$\begin{aligned}\begin{bmatrix} x_{t+1}^{(1)} \\ x_{t+1}^{(2)} \end{bmatrix} &= \begin{bmatrix} \arctan\left(\lambda x_t^{(1)} + x_t^{(2)}\right) \\ \mu x_t^{(2)} \end{bmatrix} + B v_t \\ y_t &= \sin x_t^{(1)} - \sin x_t^{(2)} + D v_t \end{aligned} \tag{20}$$

where $x_t = \begin{bmatrix} {x_t^{(1)}}^\top & {x_t^{(2)}}^\top \end{bmatrix}^\top$ is the state; $\lambda = 0.1$ and μ are the nominal model parameters; $v_t \in \mathbb{R}^3$ is normalized white Gaussian noise where B and D are the corresponding noise matrices. The initial state x_0 is Gaussian distributed with zero mean and covariance matrix $\tilde{V}_0 = 10^{-3} \cdot I_2$; x_0 is independent from v_t. In what follows, we assume that the actual model is different from the nominal one. More precisely, we consider two possible common scenarios in which model uncertainty arises:

- Case 1. The actual model parameter, say $\tilde{\mu}$, is different from the nominal one μ.
- Case 2. The actual noise process, say $\tilde{v}_t$, is different from the nominal one v_t.

Accordingly, in each case we consider a Monte Carlo experiment composed by $M = 2000$ trials to evaluate the performance of the following two smoothers:

- REKS is the robust fixed-lag extended Kalman smoother in Algorithm 1 with constant tolerance $c_t = 10^{-2}$ and based on the nominal model (20);

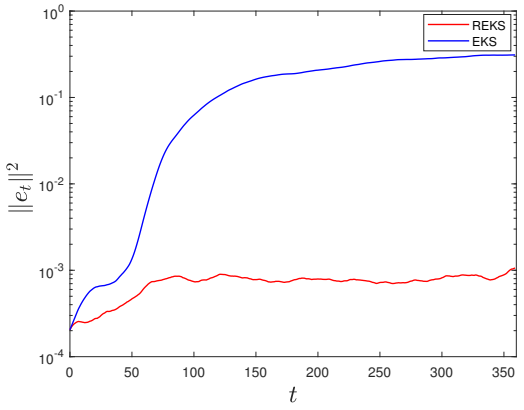


Fig. 1: Average variance of the estimation error for EKS and REKS with $L = 40$ in Case 1.

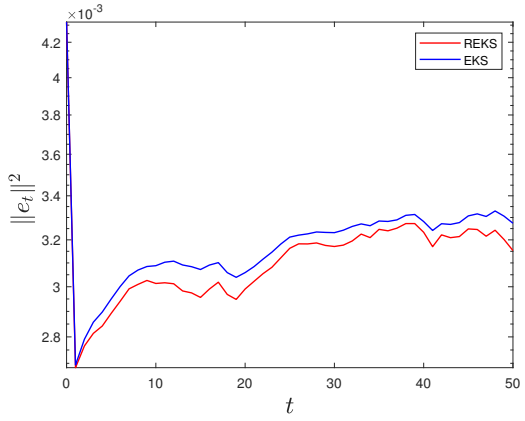


Fig. 2: Average variance of the estimation error for REKS and EKS with $L = 10$ in Case 2.

	EKS	REKS
L=5	0.1026	0.0147
L=10	0.0860	0.0073
L=20	0.1744	0.0023
L=30	0.1656	0.0015
L=40	0.1657	0.0007

TABLE I: $\bar{\sigma}^2$ for EKS and REKS with different values of lag L in Case 1.

- EKS is the standard fixed-lag extended Kalman smoother proposed in [20] and based on the nominal model (20).

More precisely, in each trial, we collect the actual state trajectory X_N and the corresponding actual measurement trajectory Y_N generated by the actual model. Then, we quantify the performance of the two smoothers through the average variance of the estimation error $\|e_t\|^2$, with $t = 1 \dots N - L$, for REKS and EKS as follows:

$$\|e_t\|^2 = \frac{1}{M} \sum_{k=1}^{M} \|x_t^k - \hat{x}_{t|t+L-1}^k\|^2 \tag{21}$$

where x_t^k is the state at time t extracted from the actual state trajectory (i.e. X_N) generated in the k-th trial, $\hat{x}_{t|t+L-1}^k$ is the corresponding smoothed estimate using EKS or REKS (i.e. both they use Y_N generated in the k-th trial). Clearly, the lower $\|e_t\|^2$ is, the better performance of the corresponding algorithm is.

Case 1: We consider the nominal model (20) with $\mu = 0.5$,

$$B = 10^{-3} \cdot \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 22.5 & 0 \end{bmatrix}, \quad D = 10^{-3} \cdot \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}.$$

The actual model is as the nominal one but the actual value of $\tilde{\mu} = 0.98$ is different from μ. Fig. 1 shows the average variance of the estimation error, i.e. $\|e_t\|^2$, for REKS and EKS obtained from the Monte Carlo experiment with $L = 40$ and $N = 400$. We see that REKS outperforms EKS. We also quantify the performance of the smoothers for different values of the lag, i.e. $L = 5$, $L = 10$, $L = 20$ and $L = 30$. Table I shows the average value of $\|e_t\|^2$ over $t = 1 \dots N - L$, namely, $\bar{\sigma}^2 = \frac{1}{N-L} \sum_{t=1}^{N-L} \|e_t\|^2$. We notice that larger L is, the better the performance of REKS is. The performance of EKS is always worse than the one of REKS and it generally deteriorates increasing L. Finally, we also considered different scenarios of parameter perturbation. For instance, we chosen as nominal parameters $\mu = 0.3$, $\lambda = 0.1$ and as actual parameters $\tilde{\mu} = 0.99$, $\tilde{\lambda} = 0.5$. Also in this case, REKS performs better than EKS.

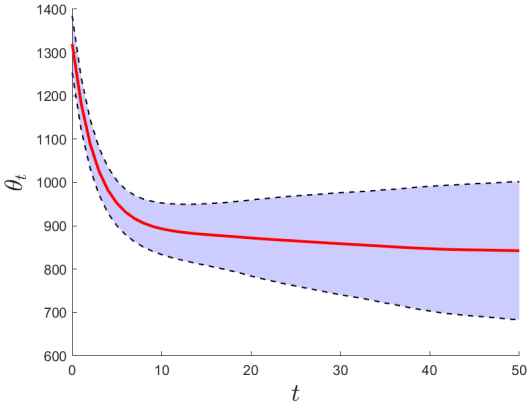


Fig. 3: Distribution over 2000 trials for the risk sensitivity parameter θ_t of REKS with $c_t = 10^{-2}$ in Case 2: the red line denotes its profile on average while the dashed lines define the corresponding confidence intervals (with level 0.95).

Case 2: We consider the nominal model (20) with $\mu = 0.98$ and the noise matrices are given by

$$B = 10^{-3} \cdot \begin{bmatrix} 0.1 & 0 & 0 \\ 0 & 4.5 & 0 \end{bmatrix}, \quad D = 10^{-5} \cdot \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}.$$

The actual model is as the nominal model but the actual noise process $\tilde{v}_t \in \mathbb{R}^3$ is colored. More precisely, $\tilde{v}_t$ is the output of the filter with transfer matrix

$$\mathcal{H}(z) = \begin{bmatrix} \frac{z-0.909}{z-0.9092} & \frac{-0.0065}{z-0.9092} & 0 \\ \frac{-0.0078z-0.0011}{z-0.9092} & \frac{1.1012z-0.9255}{z-0.9092} & \frac{-0.0006}{z-0.9092} \\ 0 & \frac{-0.0006z-0.0001}{z-0.9092} & 1 \end{bmatrix}$$

and driven by normalized white Gaussian noise. Fig. 2 shows the average variance of the estimation error, i.e. $\|e_t\|^2$, for REKS and EKS obtained from the Monte Carlo experiment with $L = 10$ and $N = 60$. We see that REKS performs better than EKS. Fig. 3 shows the distribution over all the trials of

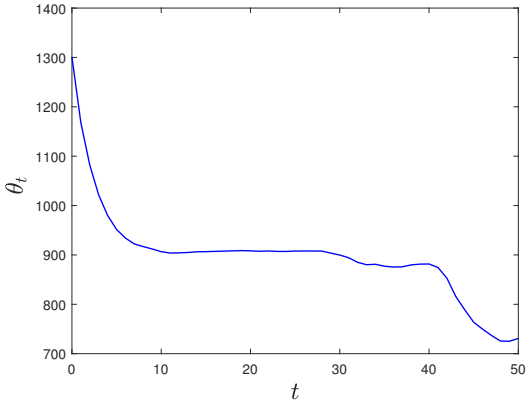


Fig. 4: One trial in the Monte Carlo experiment for the risk sensitivity parameter θ_t of REKS in Case 2.

the risk sensitivity parameter $\theta_t > 0$ of REKS: the red line denotes its profile on average, while the dashed lines define the corresponding confidence intervals (with level 0.95). It is worth noting that the evolution of the risk sensitivity parameter is different in any trial, indeed this evolution depends on the particular linearization. Finally, the confidence interval is very small at the beginning because the distribution of the initial state is the same in any trial. Fig. 4 shows evolution of the risk sensitivity parameters in one trial: although the nominal model is with constant parameters and the tolerance c_t is taken constant, θ_t does not converge to a constant value as it happens in the linear case, see [28].

V. Conclusion

In this paper, we have considered a robust nonlinear smoothing problem under model uncertainties. More precisely, we proposed a robust version of the extended fixed-lag smoother which can be used in the case that the nominal model is nonlinear and different from the actual one. There are still some questions that have not been taken into account yet and that we would like to address as a future work:

- Since the lag L of the smoother is typically large, the robust smoother proposed in Algorithm 1 is not numerically robust and efficient in terms of computational burden. The paper [16, Sec. 4] shows that there is an efficient implementation for the robust fixed-lag smoother in the case the nominal model is linear. Our conjecture is that it is possible to develop an efficient algorithm also in our setting.
- In the case the nominal model is linear, then it is possible to characterize the least favorable model over a finite simulation horizon, see [16, Sec. 5]. Therefore, one would wonder whether these ideas can be adapted in this nonlinear setting in order to characterize the least favorable model.

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